Question	Answer	Marks
5(a)		**2**

Flag	**TestNum**	**Num**	**OUTPUT**
True	7	6	
		5	
		4	
		3	
		2	
		1	7
<---------- 1 Mark ---------->			<- 1 Mark - >

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>5(b)</td><td>
<table>
<tr><th>Flag</th><th>TestNum</th><th>Num</th><th>OUTPUT</th></tr>
<tr><td>True</td><td>6</td><td>5</td><td></td></tr>
<tr><td></td><td></td><td>4</td><td></td></tr>
<tr><td></td><td></td><td>3</td><td></td></tr>
<tr><td>False</td><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td></td><td></td></tr>
<tr><td colspan="4"><----------- 1 Mark ---------->←---------1 Mark ---------- ></td></tr>
</table>
</td><td>2</td></tr>
<tr><td>5(c)</td><td>1 mark for correct purpose e.g.

Works out if the number entered is a prime number.</td><td>[1]</td></tr>
</table>

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>5(a)</td><td>
<table>
<tr><th>Height</th><th>Depth</th><th>Chlorine</th><th>OK</th><th>OUTPUT</th></tr>
<tr><td>6</td><td>2.5</td><td>2</td><td>True</td><td>Pool OK to use</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td></tr>
</table>
<table>
<tr><th>Height</th><th>Depth</th><th>Chlorine</th><th>OK</th><th>OUTPUT</th></tr>
<tr><td>4</td><td>3</td><td>1.5</td><td>True</td><td>Water too deep</td></tr>
<tr><td></td><td></td><td></td><td>False</td><td></td></tr>
</table>
<table>
<tr><th>Height</th><th>Depth</th><th>Chlorine</th><th>OK</th><th>OUTPUT</th></tr>
<tr><td>6</td><td>3.5</td><td>4</td><td>True</td><td>Water too deep</td></tr>
<tr><td></td><td></td><td></td><td>False</td><td>Too much chlorine add more water</td></tr>
</table>
1 mark for first 4 columns in each trace table (max 3)
1 mark for the output in each trace table (max 3)
</td><td>6</td></tr>
<tr><td>5(b)</td><td>Any one from:
Cannot add more water if the water is too deep
No validation e.g. allows a negative height/depth/amount of chlorine
Tells you to add chlorine when there is no water
Runs only once</td><td>1</td></tr>
</table>

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>3(a)</td><td>
<table>
<tr><th>Reject</th><th>Count</th><th>Length</th><th>OUTPUT</th></tr>
<tr><td>0</td><td>0</td><td></td><td></td></tr>
<tr><td>0</td><td>1</td><td>24.88</td><td></td></tr>
<tr><td>1</td><td>2</td><td>25.01</td><td></td></tr>
<tr><td>1</td><td>3</td><td>24.98</td><td></td></tr>
<tr><td>1</td><td>4</td><td>25.00</td><td></td></tr>
<tr><td>1</td><td>5</td><td>25.05</td><td></td></tr>
<tr><td>1</td><td>6</td><td>24.99</td><td></td></tr>
<tr><td>1</td><td>7</td><td>24.97</td><td></td></tr>
<tr><td>1</td><td>8</td><td>25.04</td><td></td></tr>
<tr><td>1</td><td>9</td><td>25.19</td><td></td></tr>
<tr><td>2</td><td>10</td><td>25.07</td><td>Batch rejected</td></tr>
<tr><td></td><td></td><td></td><td></td></tr>
</table>
One mark for each correct column max. 4
</td><td>4</td></tr>
<tr><td>3(b)(i)</td><td>• Remove <code>Length < 25.1 AND</code></td><td>1</td></tr>
<tr><td>3(b)(ii)</td><td>Two from:
• Check the reject counter after each incrementation/remove reject check after counter = 10 …
• … as soon as Reject = 2 / >1 ...
• … reject batch and end</td><td>2</td></tr>
</table>

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>4(a)</td><td>One mark for correct input columns (V, W, X, Y, Z)
One mark for correct calculation column A
One mark for correct calculation column B
One mark for correct output column

<table>
<tr><th>V</th><th>W</th><th>X</th><th>Y</th><th>Z</th><th>A</th><th>B</th><th>OUTPUT</th></tr>
<tr><td>5</td><td>4</td><td>6</td><td>2</td><td>1</td><td>56</td><td>1</td><td>Valid</td></tr>
<tr><td>9</td><td>3</td><td>2</td><td>1</td><td>6</td><td>40</td><td>7</td><td>Invalid</td></tr>
<tr><td>7</td><td>6</td><td>1</td><td>5</td><td>1</td><td>61</td><td>6</td><td>Invalid</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td></td><td></td><td></td></tr>
</table></td><td>4</td></tr>
<tr><td>4(b)</td><td>• (Use first four digits input to) calculate a check digit
• Check if the check digit input is valid</td><td>2</td></tr>
</table>